

DevOps Assignment

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git init: `git init` is used to create a new Git repository in a project folder. It initializes Git and starts tracking files in that directory.

```
aadit@DESKTOP-AADITYA MINGW64 /d/codefiles/devops/Developer-Main
● $ git init
Initialized empty Git repository in D:/codefiles/devops/Developer-Main/.git/
```

git status: `git status` shows the current state of the working directory. It tells which files are modified, staged, or untracked.

```
aadit@DESKTOP-AADITYA MINGW64 /d/codefiles/devops/Developer-Main (master)
● $ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        index.html

nothing added to commit but untracked files present (use "git add" to track)
```

git add: `git add` moves changes from the working directory to the staging area. It prepares selected files to be committed.

```
aadit@DESKTOP-AADITYA MINGW64 /d/codefiles/devops/Developer-Main (master)
● $ git add .
```

git commit: `git commit` saves the staged changes permanently in the repository. Each commit creates a snapshot of the project with a message.

```
aadit@DESKTOP-AADITYA MINGW64 /d/codefiles/devops/Developer-Main (master)
● $ git commit -m "initial commit"
[master (root-commit) 2b90b0a] initial commit
 1 file changed, 12 insertions(+)
 create mode 100644 index.html
```

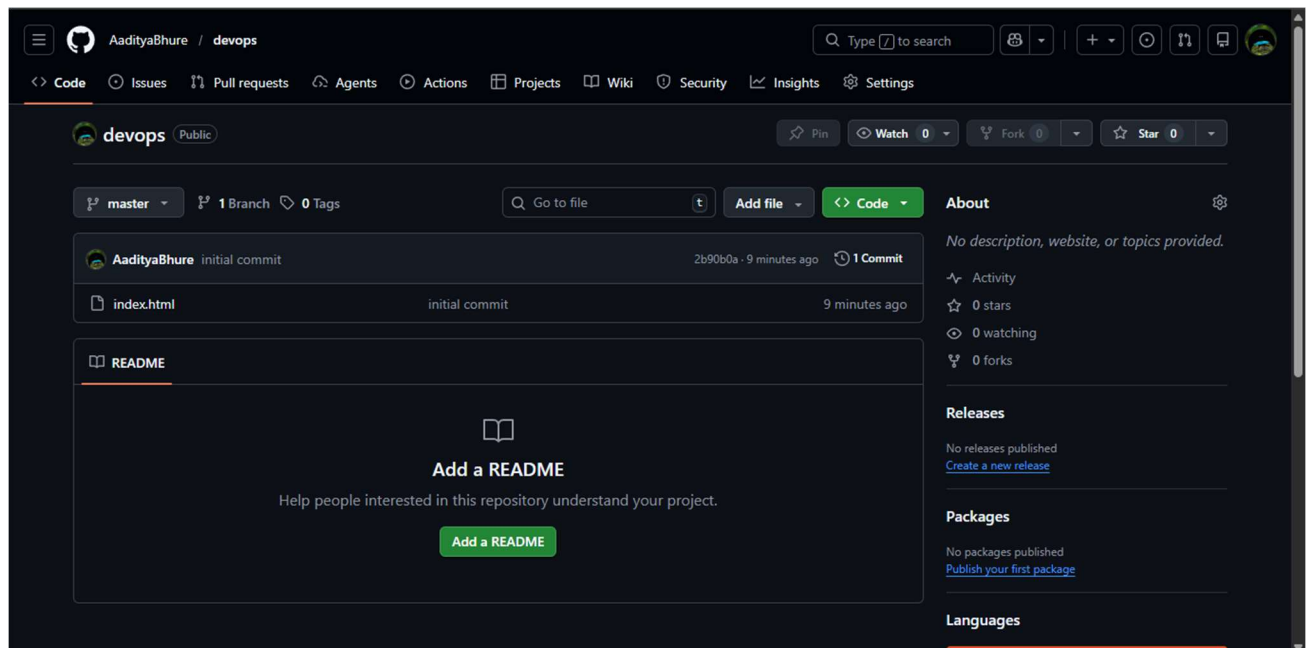
git remote: `git remote` is used to connect your local repository to a remote repository like GitHub. It allows you to push and pull code between your local system and remote server.

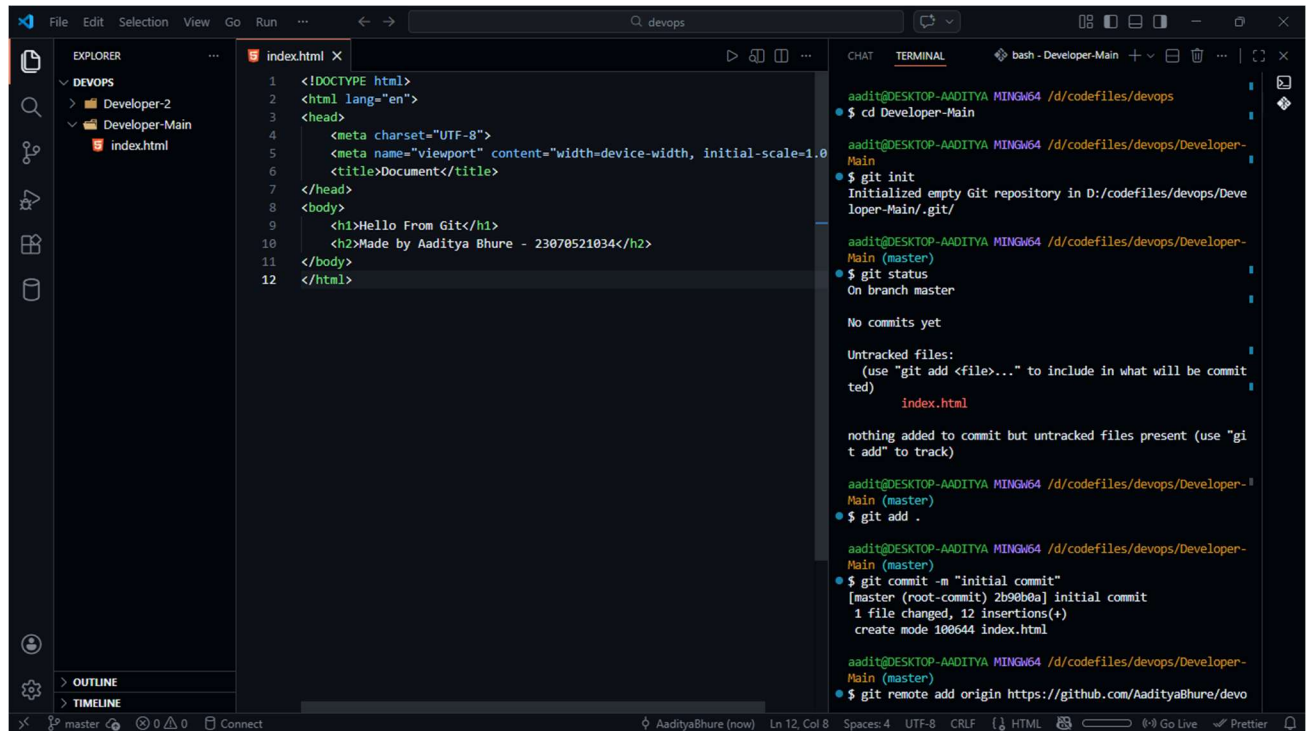
```
aadit@DESKTOP-AADITYA MINGW64 /d/codefiles/devops/Developer-Main (master)
• $ git remote add origin https://github.com/AadityaBhure/devops.git
```

git push: `git push` uploads your local committed changes to a remote repository. It updates the remote branch with your latest code.

```
aadit@DESKTOP-AADITYA MINGW64 /d/codefiles/devops/Developer-Main (master)
• $ git push origin master
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 16 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 417 bytes | 417.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/AadityaBhure/devops.git
 * [new branch]      master -> master
```

outcome:





git clone: `git clone` is used to copy a remote repository to your local system. It creates a complete working project with all files and history.

```
aadit@DESKTOP-AADITYA MINGW64 /d/codefiles/devops/Developer-2
$ git clone https://github.com/AadityaBhure/devops.git
Cloning into 'devops'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 3 (delta 0), reused 3 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.
```

git branch: `git branch` is used to create or list branches in a repository.

```
aadit@DESKTOP-AADITYA MINGW64 /d/codefiles/devops/Developer-Main (master)
$ git branch f1
```

Branches allow parallel development without affecting the main code.

git checkout: `git checkout` is used to switch between branches. It allows you to work on different versions of the project.

```
aadit@DESKTOP-AADITYA MINGW64 /d/codefiles/devops/Developer-Main (master)
• $ git checkout f1
Switched to branch 'f1'
```

git fetch: `git fetch` downloads the latest changes from the remote repository. It updates tracking branches but does not merge changes automatically.

```
aadit@DESKTOP-AADITYA MINGW64 /d/codefiles/devops/Developer-Main (f1)
• $ git fetch origin
From https://github.com/AadityaBhure/devops
* [new branch]      f1      -> origin/f1
```

git pull: `git pull` fetches and merges changes from the remote repository. It updates your local branch with the latest remote changes.

```
aadit@DESKTOP-AADITYA MINGW64 /d/codefiles/devops/Developer-Main (f1)
• $ git pull origin master
From https://github.com/AadityaBhure/devops
* branch            master    -> FETCH_HEAD
Already up to date.
```

git log: `git log` displays the commit history of the repository. It shows details like commit ID, author, date, and message.

```
aadit@DESKTOP-AADITYA MINGW64 /d/codefiles/devops/Developer-Main (f1)
• $ git log
commit 2b90b0a8d60092376ac94600a52d999b5f361606 (HEAD -> f1, origin/master,
origin/f1, origin/HEAD, master)
Author: AadityaBhure <aadityabhure001@gmail.com>
Date:   Wed Feb 18 18:53:47 2026 +0530

    initial commit
```

git merge: `git merge` combines changes from one branch into another branch. It integrates different development work into a single branch.

```
aadit@DESKTOP-AADITYA MINGW64 /d/codefiles/devops/Developer-Main (f1)
• $ git merge f1
Already up to date.
```

Outcome:

The screenshot displays a GitHub repository named 'devops' and a Visual Studio Code editor window.

GitHub Repository: devops

- Navigation bar: Code, Issues, Pull requests, Agents, Actions, Projects, Wiki, Security, Insights, Settings.
- Section: Branches (New branch button)
- Filters: Overview, Yours, Active, Stale, All.
- Search: Search branches...
- Table of branches:

Branch	Updated	Check status	Behind / Ahead	Pull request
master	21 minutes ago		Default	
f1	6 minutes ago		0 / 0	

VS Code Editor

- Explorer: Shows project structure with folders 'Developer-2' and 'Developer-Main', each containing 'index.html'.
- Editor: 'index.html' file is open, showing HTML code.
- Terminal: Shows the execution of Git commands.

index.html content:

```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, in
6   <title>Document</title>
7 </head>
8 <body>
9   <h1>Hello From Git</h1>
10  <h2>Made by Aaditya Bhure - 23070521034</h2>
11 </body>
12 </html>
```

Terminal Output:

```
bash - Developer-Main
aadit@DESKTOP-AADITYA MINGW64 /d/codefiles/devops/Developer-2 (master)
$ cd ..
aadit@DESKTOP-AADITYA MINGW64 /d/codefiles/devops (master)
$ cd Developer-Main
aadit@DESKTOP-AADITYA MINGW64 /d/codefiles/devops/Developer-Main (master)
$ git branch f1
aadit@DESKTOP-AADITYA MINGW64 /d/codefiles/devops/Developer-Main (master)
$ git checkout f1
Switched to branch 'f1'
aadit@DESKTOP-AADITYA MINGW64 /d/codefiles/devops/Developer-Main (f1)
$ git fetch origin
From https://github.com/AadityaBhure/devops
* [new branch] f1 -> origin/f1
aadit@DESKTOP-AADITYA MINGW64 /d/codefiles/devops/Developer-Main (f1)
$ git pull origin master
From https://github.com/AadityaBhure/devops
* branch master -> FETCH_HEAD
Already up to date.
aadit@DESKTOP-AADITYA MINGW64 /d/codefiles/devops/Developer-Main (f1)
$ git log
commit 2b90b0a8d60092376ac94600a52d999b5f361606 (HEAD -> f1, origin/master, origin/f1, origin/HEAD, master)
Author: AadityaBhure <aadityabhure001@gmail.com>
Date: Wed Feb 18 18:53:47 2026 +0530

    initial commit
aadit@DESKTOP-AADITYA MINGW64 /d/codefiles/devops/Developer-Main (f1)
$ git merge f1
Already up to date.
aadit@DESKTOP-AADITYA MINGW64 /d/codefiles/devops/Developer-Main (f1)
$
```